Example 3.34 Express each of the following sum or difference as a product

(i) $\sin 50^\circ + \sin 20^\circ$ (ii) $\cos 6\theta + \cos 2\theta$ (iii) $\cos \frac{3x}{2} - \cos \frac{9x}{2}$.

Solution:

(i) We know that $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$

Take $C = 50^\circ$ and $D = 20^\circ$. We have

$$\sin 50^\circ + \sin 20^\circ = 2 \sin \frac{50^\circ + 20^\circ}{2} \cos \frac{50^\circ - 20^\circ}{2} = 2 \sin 35^\circ \cos 15^\circ$$

(ii) We know that $\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$

Take $C = 6\theta$ and $D = 2\theta$. We have

$$\cos 6\theta + \cos 2\theta = 2 \cos \frac{6\theta + 2\theta}{2} \cos \frac{6\theta - 2\theta}{2} = 2 \cos 4\theta \cos 2\theta$$

(iii) We know that $\cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \frac{D-C}{2}$

Take $C = \frac{3x}{2}$ and $D = \frac{9x}{2}$. We have

$$\cos \frac{3x}{2} - \cos \frac{9x}{2} = 2 \sin \frac{\frac{3x}{2} + \frac{9x}{2}}{2} \sin \frac{\frac{9x}{2} - \frac{3x}{2}}{2} = 2 \sin 3x \sin \frac{3x}{2}.$$

Example 3.35 Find the value of $\sin 34^\circ + \cos 64^\circ - \cos 4^\circ$.

Solution:

$$\begin{aligned} \text{We have } \sin 34^\circ + \cos 64^\circ - \cos 4^\circ &= \sin 34^\circ - 2 \sin \left(\frac{64^\circ + 4^\circ}{2} \right) \sin \left(\frac{64^\circ - 4^\circ}{2} \right) \\ &= \sin 34^\circ - 2 \sin 34^\circ \sin 30^\circ = 0. \end{aligned}$$

Example 3.36 Show that $\cos 36^\circ \cos 72^\circ \cos 108^\circ \cos 144^\circ = \frac{1}{16}$.

Solution:

$$\begin{aligned} \text{L.H.S.} &= \cos 36^\circ \cos (90^\circ - 18^\circ) \cos (90^\circ + 18^\circ) \cos (180^\circ - 36^\circ) \\ &= \sin^2 18^\circ \cos^2 36^\circ \\ &= \left(\frac{\sqrt{5} - 1}{4} \right)^2 \left(\frac{\sqrt{5} + 1}{4} \right)^2 = \frac{1}{16}. \end{aligned}$$